The Durbin Amendment and Bank Deposit Fee Adjustments: A Regression Discontinuity Analysis at the $10 Billion Threshold

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Abstract

In the U.S. debit card market, large banks lost substantial interchange revenue after the Durbin Amendment capped fees for issuers above $10 billion. However, whether banks shifted these losses onto consumers through higher deposit fees remains unclear. To address this question, this paper constructs a bank-level panel from FFIEC Call Reports (2009–2013) and estimates a sharp regression discontinuity design (RDD) around the $10 billion threshold. Across all specifications, the analysis finds no statistically significant discontinuity in deposit fee growth at the cutoff. Diagnostic tests reveal sparse data density near the threshold and sample imbalance. These results suggest that, with Call Report aggregates and few mid-sized banks, the Durbin threshold does not yield a powerful local experiment, highlighting the data requirements for RDDs in banking regulation.

JEL Codes: G21, G28, L11

I. Introduction

In July 2010, the U.S. Congress passed the Dodd–Frank Wall Street Reform and Consumer Protection Act[1]. Section 1075, known as the Durbin Amendment, directed the Federal Reserve to cap the interchange fees that large banks[2] could charge merchants for processing debit card transactions (Kay, Manuszak, and Vojtech, 2018). The resulting rule, Regulation II, established a sharp threshold: only banks with consolidated assets of $10 billion or more became subject to the cap, while smaller banks remained exempt (Manuszak and Wozniak, 2017). When the rule took effect on October 1, 2011, the average interchange fee for covered banks fell from approximately $0.44 to $0.21 per transaction (Kay, Manuszak, and Vojtech, 2014). Evans, Chang, and Joyce (2015) estimate that this revenue shock reduced covered banks' interchange revenue by roughly $8 billion per year.

From an economic perspective, this policy shock provides a useful setting for analyzing the pricing behavior of multi-product firms under imperfect competition. Commercial banks sell bundles of products: debit card services, checking accounts, overdraft protection, and other retail services to relatively sticky customers. A firm with market power chooses a vector of prices across these products to maximize profit subject to demand and regulatory constraints. When a binding cap reduces the margin on one product, a profit-maximizing multi-product bank may respond by adjusting prices on other products in the bundle, such as monthly maintenance fees or overdraft charges, as long as customers respond to the overall package rather than to each fee in isolation (Armstrong, 2006; Rochet and Tirole, 2003).

Applied to Durbin, this logic implies that a cap on interchange fees can lead to higher prices in seemingly separate markets, such as deposit account fees. Even though deposit service charges and interchange fees are recorded in different income categories, they both contribute to the same profit function. When interchange revenue falls, banks may respond by raising monthly maintenance fees, tightening minimum balance requirements, reducing rewards, or increasing overdraft and other account charges. This "waterbed effect"[3] does not require a formal two-sided market model; it follows directly from standard pricing under imperfect competition for a bank that can reallocate markups across its product portfolio.

Existing empirical work on the Durbin Amendment supports this margin-shifting story. Using regulatory Call Reports[4] and holding-company data in a difference-in-differences framework, Kay, Manuszak, and Vojtech (2018) estimate that covered banks recouped over 90 percent of their lost interchange revenue by increasing service charges on deposit accounts. Using more granular branch-

[1]The full text of the Durbin Amendment is available on page 2068 of Government Printing Office (2010). The Board's press release and the text for the proposed and final regulations are available at Federal Reserve Board (2010) and Federal Reserve Board (2011a), respectively.

[2]This paper uses the term "banks" to refer to banks, savings and loan associations, and credit unions. All of these institutions typically issue debit cards as part of demand depository accounts they offer consumers and small business customers; see Evans, Chang, and Joyce (2015, p. 2).

[3]Kay, Manuszak, and Vojtech (2014) note that a similar phenomenon has been studied in telecommunications markets, where changes in interconnection fees can lead to offsetting price adjustments for subscribers, a pattern often labeled the "waterbed effect".

[4]Specific information on the FFIEC Call Reports is available at the FFIEC reporting forms website.

level account data, Mukharlyamov and Sarin (2025) document large reductions in the availability of free checking and sizeable increases in monthly account fees at covered institutions. At the same time, survey and pricing evidence suggests that merchants passed little of their interchange savings through to final retail prices, implying that consumers bore most of the incidence of the reform (Wang, Schwartz, and Mitchell, 2014; Haltom and Wang, 2015; Evans, Chang, and Joyce, 2015).

However, much of this literature relies on size-based difference-in-differences designs that compare all banks above $10 billion to all banks below $10 billion. This strategy requires a parallel trends assumption: in the absence of Durbin, fee income at large and small banks would have evolved similarly. That assumption is strong. Large banks and small community banks differ systematically in observable characteristics such as asset size, lending intensity, and business models, as well as in their exposure to other post-crisis regulations. In the Call Report sample used here, treated banks have mean assets of roughly $100 billion, while control banks average under $0.5 billion. These stark differences raise concerns about using the entire size distribution to construct a credible difference-in-differences counterfactual.

This paper instead focuses on a more local question: did banks just above the $10 billion Durbin threshold increase deposit service charge income more than otherwise similar banks just below the threshold after the cap took effect? To answer this, the analysis exploits the regulatory cutoff directly and estimates a sharp regression discontinuity design centered at the $10 billion asset threshold that determines Durbin coverage. Under standard RDD assumptions, banks just above and just below the threshold should be similar in all relevant respects except their regulatory status, so any discrete jump in fee changes at the cutoff can be attributed to Durbin coverage.

The analysis proceeds in three steps. First, it constructs a bank-level panel from FFIEC Call Reports merged with the Federal Reserve's lists of debit-card issuers covered by and exempt from Regulation II. It then defines a change-in-log-fees outcome by averaging log service charge income in pre- and post-policy periods and taking their difference. Second, it estimates local linear RDDs of this outcome on a running variable defined as 2011:Q2 total assets minus $10 billion, with and without a pre-policy loans-to-assets covariate. Third, it assesses bandwidth sensitivity and runs standard diagnostics, including covariate balance tests and a density test for manipulation of the running variable.

Despite this stronger identification strategy, the paper finds no statistically significant discontinuity in deposit fee growth at the $10 billion threshold. Point estimates are positive but imprecise across bandwidths and specifications. Diagnostic tests help explain why. First, the analytic sample is overwhelmingly composed of very small banks: out of more than 3,900 institutions, only a few dozen lie above $10 billion, and only a handful fall within a narrow window around the cutoff. Second, both the covariate-balance test and a McCrary-style density assessment indicate that the data are sparse and somewhat unbalanced near the threshold. Together, these features limit the power of the RDD and weaken its internal validity.

Accordingly, the main contribution of this paper is methodological rather than substantive. Taken

on its own, the analysis cannot overturn the stronger evidence from account-level and bank-holding-company data that banks shifted costs onto depositors after Durbin. Instead, it illustrates how a seemingly ideal policy cutoff can fail to deliver precise or convincing RDD estimates when the available data provide too few observations near the threshold and when aggregate outcome measures bundle many underlying price margins.

The rest of the paper is organized as follows. Section II describes the data and empirical strategy, including the definition of the outcome, the running variable, and the RDD specification and diagnostics. Section III presents the results, including a complementary event study and the main RDD estimates. Section IV discusses interpretation, links back to the theoretical framework and prior literature, outlines the study's limitations, and concludes.

II. Empirical Approach

II.1 Data Sources

This paper uses supervisory data from the Federal Financial Institutions Examination Council (FFIEC) Consolidated Reports of Condition and Income (Call Reports)[5] for U.S. commercial banks from 2009:Q1 through 2013:Q4. Call Reports provide quarterly balance sheet and income statement information for all insured commercial banks and are the standard source in the Durbin literature (Kay, Manuszak, and Vojtech, 2018; Manuszak and Wozniak, 2017; Mukharlyamov and Sarin, 2025). The two key items for this study are total assets (RCFD2170), which define Durbin coverage and the running variable, and service charges on deposit accounts (RIAD4080), which measure fee income charged directly to depositors.

Because Durbin applies only to debit-card-issuing institutions, the sample is restricted to banks that appear on the Federal Reserve Board's lists of issuers covered by and exempt from Regulation II[6]. To address missing asset values in the critical assignment quarter (2011:Q2), the Call Reports are merged with FDIC BankFind Suite data[7] and an FDIC institution file that cross-walks CERT numbers and IDRSSD identifiers[8]. When 2011:Q2 assets are missing in Call Reports but available in BankFind, the BankFind value is used. Banks with unresolved missing assets in 2011:Q2 are dropped so that treatment status is always well defined. This pre-policy asset measure is standard in the Durbin literature and avoids classifying as treated any banks that might have adjusted their assets in response to the regulation (Kay, Manuszak, and Vojtech, 2018).

The raw panel contains roughly 79,000 bank-quarter observations. For the RDD, the panel is

[5] Bulk Call Report data are downloaded from the FFIEC bulk data portal.

[6] The Federal Reserve's Regulation II issuer lists identify banks subject to the interchange cap (assets of at least $10 billion) and exempt banks (below $10 billion).

[7] The FDIC BankFind Suite provides institution-level financial reporting tables, historical asset series, and metadata for all FDIC-insured banks.

[8] The FDIC institution file provides the mapping between FDIC CERT numbers and Federal Reserve IDRSSD numbers, along with headquarter location, charter class, and demographic attributes.

collapsed to a bank-level cross-section. The final analytic sample keeps banks that have valid service charge data in at least one pre-policy quarter and at least one post-policy quarter, and that also have a valid measure of 2011:Q2 total assets. The resulting sample contains 3,905 banks, of which 42 are above the $10 billion Durbin threshold and 3,863 are below.

Table 1 presents summary statistics for the analytic sample. The data reveal a profound structural skew in the U.S. banking sector. The 42 treated banks differ from the 3,863 control banks by orders of magnitude, with mean assets of $124 billion versus $0.37 billion. This extreme heterogeneity underscores the need for an RDD approach, which restricts comparison to the local threshold, rather than a difference-in-differences design that would require the implausible assumption that megabanks and community banks share parallel trends. The table also reports the change in log service charge income and a pre-policy loans-to-assets ratio, which is later used as a covariate in the RDD.

Table 1: Descriptive Statistics for RDD Analytic Sample (by Durbin Coverage)

	Mean		Standard deviation	
	Control banks (assets < $10B)	Durbin-covered banks (assets ≥ $10B)	Control banks (assets < $10B)	Durbin-covered banks (assets ≥ $10B)
Number of banks	3,863	42		
Total assets in 2011:Q2* (Call Report total assets)	371,855	124,234,475	801,369	324,573,877
Average log fee income, pre-policy[a]	5.0	10.0	1.0	2.0
Average log fee income, post-policy[b]	5.0	11.0	1.0	2.0
Change in log fee income (post − pre)[c]	0.00	0.00	0.00	1.00
Loans-to-assets ratio	0.60	0.55		
All banks (pooled)				
Number of banks	3,905			
Total assets in 2011:Q2	1,704,052		35,641,041	
Average log fee income, pre-policy	5.0		2.0	
Average log fee income, post-policy	5.0		2.0	
Change in log fee income (post − pre)	0.00		0.00	
Loans-to-assets ratio	0.60			

Notes: *Assets are in thousands of dollars. "Durbin-covered" banks have consolidated assets at or above $10 billion as of 2011:Q2; "control" banks are below $10 billion. Total assets are measured using the Call Report total assets item in 2011:Q2. [a]Average of log quarterly service charge income over the pre-policy window (2009:Q1–2011:Q1). [b]Average over the post-policy window (2012:Q1–2013:Q4). [c]Difference between post- and pre-policy averages for each bank. Loans-to-assets ratios are reported as means only. All statistics are computed on the 3,905-bank RDD analytic sample.

In addition to Table 1, the paper uses two descriptive figures as diagnostics: a histogram of assets relative to the $10 billion cutoff, which shows the extreme skewness of the size distribution and the sparsity of banks near the threshold, and an RDD-style scatterplot of the outcome against the running variable with binned means and fitted polynomials.

II.2 Econometric Model

Outcome: Change in Log Service Charge Income

Durbin's interchange fee cap is a permanent shock to one component of banks' revenue. Theory on multi-product pricing under imperfect competition predicts that affected banks may adjust the path of deposit fees over time to restore their target rate of return (Armstrong, 2006; Kay, Manuszak, and Vojtech, 2018). To capture such adjustments while netting out banks' different starting levels of fee income, the outcome is defined as the change in log service charge income between a pre-policy baseline and a post-policy period.

Let F_{it} denote quarterly service charge income for bank i in quarter t. Because fee income tends to scale roughly in proportion to bank size, the specification assumes a multiplicative relationship between fees and bank scale rather than a purely linear one. The analysis therefore uses the log-transformed variable:

$$y_{it} = \ln(F_{it}), \tag{1}$$

so that differences in y_{it} can be interpreted as approximate percentage changes in fee income.

For the RDD, the panel is collapsed to one observation per bank by averaging over time. Define the pre-policy baseline period as

$$\tau_{pre} = \{2009{:}Q1, \ldots, 2011{:}Q1\}, \tag{2}$$

and the post-policy period as

$$\tau_{post} = \{2012{:}Q1, \ldots, 2013{:}Q4\}. \tag{3}$$

Then compute bank-specific averages of log service charge income in each window:

$$\bar{y}_i^{pre} = \frac{1}{|\tau_{pre}|} \sum_{t \in \tau_{pre}} y_{it}, \tag{4}$$

$$\bar{y}_i^{post} = \frac{1}{|\tau_{post}|} \sum_{t \in \tau_{post}} y_{it}. \tag{5}$$

The RDD outcome is the difference between these two averages:

$$Y_i \equiv \Delta \ln(F_i) = \bar{y}_i^{post} - \bar{y}_i^{pre}. \tag{6}$$

Y_i measures the change in a typical quarter's percentage-scaled fee income for bank i between the pre- and post-Durbin periods. Each bank therefore contributes a single outcome Y_i and a single running-variable value, making the RDD analysis purely cross-sectional.

The use of averages rather than cumulative sums follows the logic of the Durbin literature for three reasons. First, averaging makes the measure robust to occasional missing quarters: a bank with

one missing report is not mechanically treated as having lower total fees. Second, averaging makes estimates comparable across robustness checks that vary the length of the time window. Third, averaging smooths quarter-specific seasonal volatility, so the cross-sectional RDD focuses on the persistent shift in fee policy rather than transitory noise.

To address anticipation effects, the quarters immediately surrounding implementation are excluded from the baseline. The legislative debate and rulemaking process were well publicized before the October 2011 effective date, and the event-study evidence later in the paper shows elevated fee growth for large banks in 2011:Q2–2011:Q3. The analysis therefore drops 2011:Q2–2011:Q3 from the pre-policy window and begins the post-policy window in 2012:Q1. This construction is intended to provide a cleaner pre-treatment benchmark that is not contaminated by preemptive pricing adjustments (Lee and Lemieux, 2010; Angrist and Pischke, 2009).

Running Variable, Treatment, and Covariate

The Durbin Amendment applies deterministically to banks with consolidated assets of at least \$10 billion. Following the existing literature, assets are measured in 2011:Q2, after Dodd–Frank's passage but before implementation of Regulation II, so that treatment status is predetermined and not itself a response to the cap.

Let A_i denote bank i's total assets in 2011:Q2, expressed in billions of dollars. The running variable is defined as

$$X_i = A_i - 10, \tag{7}$$

so that $X_i = 0$ corresponds to the \$10 billion threshold. The treatment indicator is

$$D_i = \mathbf{1}\{X_i \geq 0\}, \tag{8}$$

equal to one for banks at or above \$10 billion and zero otherwise.

As a pre-policy covariate, the analysis also constructs a loans-to-assets ratio for 2011:Q2:

$$LTA_i = \left.\frac{\text{Total Loans}_i}{\text{Total Assets}_i}\right|_{2011:Q2}. \tag{9}$$

This variable summarizes each bank's business model and risk exposure before Durbin. Prior RDD guidance recommends using such pre-treatment covariates both as balance tests and as precision-improving controls. In this paper, LTA_i is used in a covariate-balance regression and in a covariate-adjusted RDD specification.

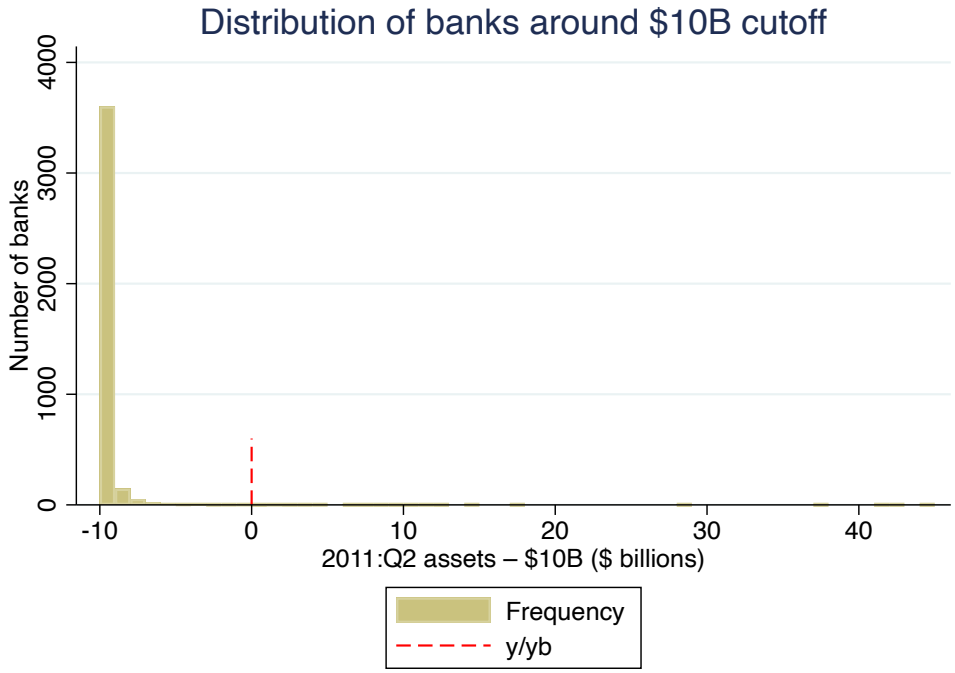

Figure 1: Distribution of banks around the $10 billion asset cutoff. Histogram of 2011:Q2 total assets (centered at $10 billion) for banks in the RDD analytic sample.

RDD Identification and Specification

The Durbin rule generates a sharp RDD: conditional on 2011:Q2 assets, the probability of treatment jumps from 0 to 1 at $X_i = 0$. Under the standard RDD assumption that banks just above and below the threshold would have followed smoothly evolving fee paths in the absence of the policy, any discrete jump in Y_i at $X_i = 0$ can be interpreted as the causal effect of coverage for banks near the cutoff.

Let $Y_i(0)$ and $Y_i(1)$ denote potential outcomes for the change in log fee income without and with Durbin coverage. The key identifying assumption is continuity of the untreated potential outcome at the threshold:

$$\lim_{x \downarrow 0} E[Y_i(0) \mid X_i = x] = \lim_{x \uparrow 0} E[Y_i(0) \mid X_i = x]. \tag{10}$$

Intuitively, in the absence of Durbin, banks just above and just below $10 billion would have followed smoothly evolving fee paths. Any discrete jump in $E[Y_i \mid X_i]$ at $X_i = 0$ can then be attributed to the policy. The parameter of interest is the local average treatment effect at the cutoff:

$$\tau = \lim_{x \downarrow 0} E[Y_i(1) \mid X_i = x] - \lim_{x \uparrow 0} E[Y_i(0) \mid X_i = x]. \tag{11}$$

Following standard RDD practice, the main specification is a local linear regression with potentially

different slopes on either side of the threshold:

$$Y_i = \alpha + \tau D_i + \beta_0 X_i + \beta_1 (D_i \times X_i) + \varepsilon_i. \tag{12}$$

Here, τ measures the jump in the change in log fee income at \$10 billion, β_0 captures the relationship between bank size and fee changes for banks below the cutoff, and β_1 allows this relationship to differ above the cutoff.

The paper also estimates a covariate-adjusted RDD that adds the pre-policy loans-to-assets ratio:

$$Y_i = \alpha + \tau D_i + \beta_0 X_i + \beta_1 (D_i \times X_i) + \theta LTA_i + \varepsilon_i. \tag{13}$$

Including LTA_i does not change identification, but it may reduce residual variance and tighten confidence intervals when lending intensity is correlated with later fee adjustments. All RDD estimates are reported with heteroskedasticity-robust (HC1) standard errors.

Bandwidth Choice and Diagnostic Tests

A central decision in RDD is the bandwidth, or the range around the cutoff used for estimation. Narrow bandwidths improve local comparability but reduce sample size and statistical power. Wider bandwidths increase power but risk bias if the outcome–size relationship is nonlinear.

In the main RDD specifications, the paper initially attempts to use data-driven bandwidth selection implemented through the local polynomial methods of Calonico, Cattaneo, and Farrell (2020). In principle, these methods choose an optimal symmetric or asymmetric window around the cutoff based on mean-squared-error criteria. In this dataset, however, the algorithm does not converge to a feasible optimal bandwidth. This failure is itself informative and reflects the extreme sparsity of treated banks near the threshold. There are simply too few observations on the right side of the cutoff for the automatic procedure to construct a stable local window.

Given this limitation, the analysis adopts a conservative manual choice and reports baseline RDD estimates using a wide bandwidth of ±\$50 billion, measured in billions of dollars of 2011:Q2 assets. To assess robustness and connect to the existing Durbin literature, it also reports estimates using narrower symmetric and asymmetric windows such as ±\$5 billion, ±\$7 billion, and a −\$4 billion to +\$15 billion window.

Following standard RDD practice, two sets of diagnostic tests are implemented. First, the paper estimates a covariate-balance regression using the pre-policy loans-to-assets ratio as the outcome:

$$LTA_i = \alpha + \delta D_i + \beta_0 X_i + \beta_1 (D_i \times X_i) + u_i. \tag{14}$$

If banks just above and below \$10 billion are comparable, the coefficient on the treatment indicator, δ, should be small and statistically insignificant.

Second, the paper examines the density of the running variable near zero to assess whether banks manipulated assets to avoid Durbin coverage. The main evidence is visual, based on the histogram of 2011:Q2 assets with a vertical line at $10 billion. If the RDD is valid, the density of banks should be continuous across the threshold. The histogram is extremely right-skewed, with most banks far below $10 billion, but it does not show visible bunching just below the threshold. This supports the no-manipulation assumption, although it also highlights the central limitation of the data: the density of banks in the neighborhood of $10 billion is very low.

II.3 Complementary Event-Study Specification

Although the RDD is the paper's primary identification strategy, it also estimates a two-way fixed-effects event study using the full bank-quarter panel as complementary evidence on timing and pre-trends:

$$\ln(F_{it}) = \alpha_i + \lambda_t + \sum_{k \neq -1} \gamma_k \big(\mathbf{1}\{\text{rel_time}_{it} = k\} \times LargeBank_i\big) + \varepsilon_{it}, \tag{15}$$

where α_i are bank fixed effects, λ_t are quarter fixed effects, and rel_time_{it} indexes quarters relative to 2011:Q4, with 2011:Q3 ($k = -1$) as the omitted reference period. Standard errors are clustered at the bank level.

This specification is not the main causal estimator because it compares all large banks to all small banks and therefore relies on a parallel trends assumption that appears violated in the data. Instead, the event study is used for two narrower purposes. First, it documents pre-trends and possible anticipation effects around implementation. Second, it provides evidence that helps motivate the construction of the pre- and post-policy windows used to define Y_i.

III. Results

III.1 Event-Study Evidence

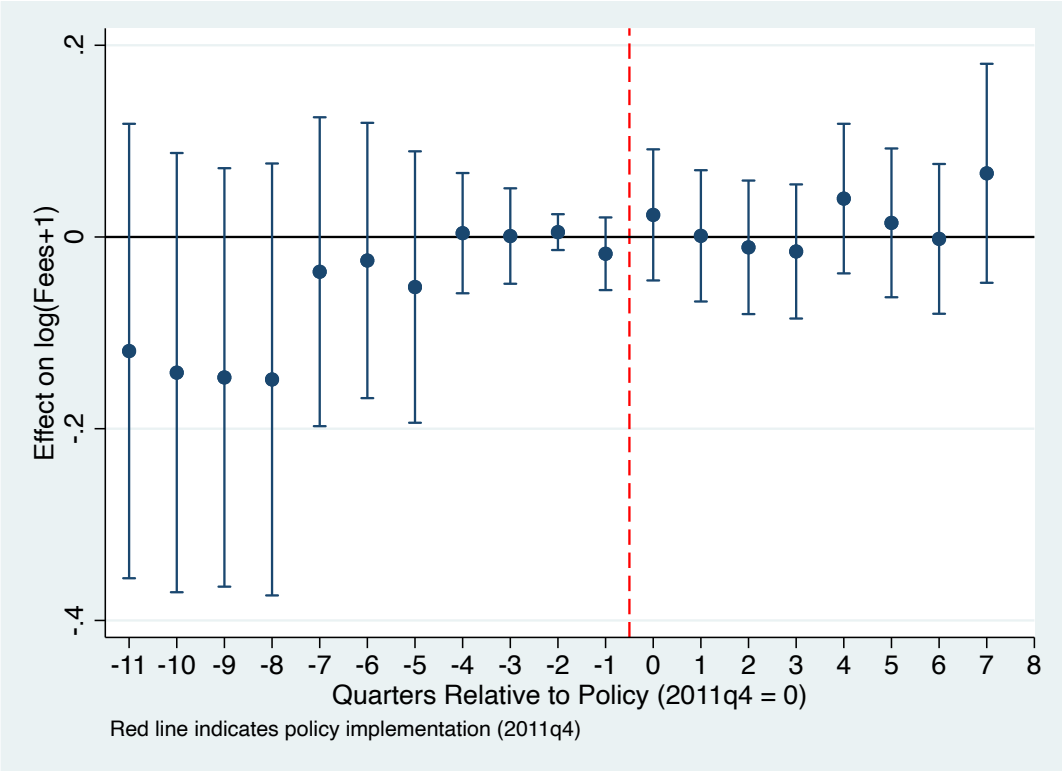

Figure 2: Event study plot of log service charge income.

Before collapsing the panel to a cross-section, the paper estimates the event-study specification described above. The regression compares quarterly log service charge income for large banks (assets above \$10 billion) to smaller banks over 2009:Q1–2013:Q4. It includes bank fixed effects, quarter fixed effects, and a full set of leads and lags of a Durbin-coverage dummy interacted with quarter-relative-to-implementation dummies. The omitted period is 2011:Q3, one quarter before the October 2011 effective date.

Figure 2 plots the event-study coefficients with 95 percent confidence intervals for 11 pre-policy quarters ($k = -11$ to $k = -2$) and 8 post-policy quarters ($k = 0$ to $k = 8$). The pre-policy coefficients fluctuate substantially, and several are statistically significant. In particular, the last two pre-Durbin quarters are both positive and statistically significant, indicating that large banks were already raising fees faster than small banks before the cap took effect. A joint F-test of all pre-policy coefficients rejects the null of no pre-trend:

$$F(10, 3967) = 2.04, \qquad p = 0.0261. \tag{16}$$

This rejection means the standard difference-in-differences parallel trends assumption does not hold

in this dataset. The pattern is consistent with anticipation effects: once the Durbin cap and its implementation timeline were known, large banks may have begun raising fees ahead of the effective date.

Post-policy coefficients are mixed rather than showing a smooth, sustained jump. The first post-policy quarter shows a positive spike of roughly 0.19 log points, but subsequent quarters alternate between small positive and negative coefficients, and several are statistically insignificant. Overall, the event study suggests that large banks' fee trajectories diverged from small banks' before the regulation took effect. This motivates moving away from a difference-in-differences design that relies on parallel trends and toward the cross-sectional RDD that focuses only on banks near the $10 billion cutoff.

At the same time, if large banks act as price leaders, any initial increases in their deposit fees might induce smaller banks to follow. Such spillover behavior would compress the gap between treated and control banks, making it harder for either the event study or the RDD to detect the full industry-wide waterbed effect in aggregate Call Report data.

III.2 Main RDD Estimates

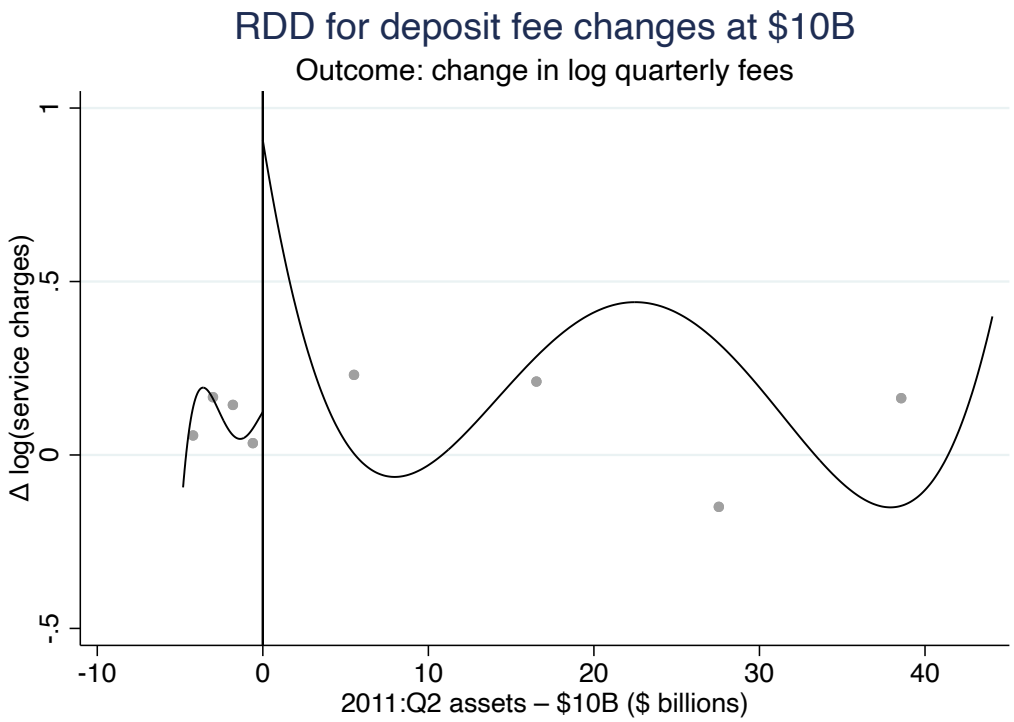

Figure 3: Regression discontinuity in deposit fee changes at the $10 billion cutoff. Binned scatter of change in average log quarterly service charge income (post–pre) versus 2011:Q2 assets relative to $10 billion, with separate local polynomial fits on each side.

Before estimating the main RDD, the paper first attempts to use a standard data-driven bandwidth selector based on the mean-squared-error procedure of Calonico, Cattaneo, and Farrell (2020) to choose the local window around the \$10 billion cutoff. In this dataset, the procedure does not return a feasible optimal bandwidth, which is consistent with the extreme sparsity of treated banks near the threshold. The baseline RDD estimates therefore use a wide bandwidth of ±\$50 billion, followed by narrower windows in a sensitivity analysis.

Table 2: Regression Discontinuity Estimates for Changes in Deposit Fee Income

	Baseline RDD	RDD with pre-policy loans-to-assets
RD estimate at \$10B cutoff[a]	0.206	0.423
Robust z-statistic	1.27	1.58
p-value	0.203	0.114
95% confidence interval	$[-0.202,\ 0.952]$	$[-0.178,\ 1.667]$
Bandwidth (in \$ billions)[b]	50	50
Observations (banks)	3,905	3,851
Effective obs. below cutoff	3,863	3,837
Effective obs. above cutoff	28	13
Pre-policy loans-to-assets included?	No	Yes

Notes: Outcome is the change in average log quarterly service charge income between the pre-policy (2009:Q1–2011:Q1) and post-policy (2012:Q1–2013:Q4) windows for each bank. [a]Local average treatment effect of Durbin coverage at the \$10 billion asset threshold, estimated using a sharp RDD with local linear polynomials and a triangular kernel. [b]Half-width of the window around the cutoff for the running variable, defined as total assets in 2011:Q2 (in billions of dollars) minus \$10 billion. In both columns, heteroskedasticity-robust (HC1) standard errors are used.

Table 2 reports the baseline regression discontinuity estimates for the change in average log quarterly service charge income between the pre-Durbin period (2009:Q1–2011:Q1) and the post-Durbin period (2012:Q1–2013:Q4). The running variable is 2011:Q2 total assets, measured in billions of dollars and centered at \$10 billion. The treatment indicator equals one for banks at or above the threshold.

In the baseline RDD, the estimated jump in fee growth at the \$10 billion cutoff is

$$\hat{\tau} = 0.206 \text{ log points}, \tag{17}$$

with a robust z-statistic of 1.27 and a p-value of 0.203. Economically, a 0.206 log-point increase corresponds to roughly 23 percent higher quarterly service charge income for banks just above the threshold relative to those just below. However, the 95 percent confidence interval is wide, ranging from -0.202 to 0.952 log points. In percentage terms, the data are consistent with anything from about an 18 percent decline to more than a 150 percent increase in fee income. Figure 3 shows a modest upward step at \$10 billion but very wide confidence bands, reflecting the limited number of treated banks near the threshold.

The covariate-adjusted specification raises the estimated treatment effect to

$$\hat{\tau} = 0.423 \text{ log points}, \tag{18}$$

which corresponds to roughly 50 percent higher quarterly fee income for treated banks at the cutoff. The associated z-statistic is 1.58 and the p-value is 0.114, so the estimate remains statistically indistinguishable from zero at conventional significance levels. The 95 percent confidence interval still covers a wide range of economically important effects, from a small decline to more than a 400 percent increase in fee income.

The loans-to-assets covariate is positively related to fee growth: banks with higher loan intensity before Durbin tended to increase deposit fees more afterward. Including this covariate slightly improves precision by absorbing some outcome variation, but it does not change the basic conclusion. Both specifications are consistent with the large waterbed effects found in prior work, but the confidence intervals are too wide to pin down the magnitude in this dataset.

Overall, the main RDD estimates point toward higher post-Durbin service charge income for banks above \$10 billion, but the evidence is weak and noisy. With so few treated banks close to the cutoff, the design does not have enough power to distinguish between modest and very large effects.

III.3 Bandwidth Sensitivity

Table 3: Bandwidth Sensitivity of RDD Estimates for Deposit Fee Changes

	Bandwidth ±\$5B	Bandwidth ±\$7B	Bandwidth −\$4B to +\$15B
	($h = 5$)	($h = 7$)	($h_L = 4$, $h_R = 15$)
RD estimate at \$10B cutoff	0.915	0.786	0.497
Robust z-statistic	0.78	1.03	1.48
p-value	0.435	0.302	0.140
95% confidence interval	[−1.274, 2.961]	[−0.926, 2.990]	[−0.271, 1.928]
Bandwidth (lower, upper in \$B)[a]	[−5, 5]	[−7, 7]	[−4, 15]
Observations (banks)	3,905	3,905	3,905
Effective obs. below cutoff	29	65	22
Effective obs. above cutoff	12	13	22

Notes: Outcome is the change in average log quarterly service charge income (post minus pre) for each bank. Each column reports sharp RDD estimates from `rdrobust` with a different manual bandwidth choice. All specifications use local linear polynomials, a triangular kernel, and HC1 robust variance estimation. [a]Bandwidths are expressed in terms of the running variable (2011:Q2 total assets minus \$10 billion, in billions of dollars).

Table 3 examines how the estimated treatment effect changes when attention is restricted to narrower windows around the \$10 billion threshold. Each column reports the same local linear RDD as in the main specification, but with a different bandwidth for the assets running variable: ±\$5 billion, ±\$7 billion, and an asymmetric window from −\$4 billion to +\$15 billion.

Across all three bandwidths, the point estimates are positive and economically large. Taken literally, these correspond to roughly 150 to 250 percent higher fee income for treated banks in the narrow symmetric windows and about 65 percent higher fee income in the asymmetric window. However, none of these estimates is statistically significant. The p-values are 0.435, 0.302, and 0.140, respectively, and the 95 percent confidence intervals are extremely wide. For example, in the ±\$5

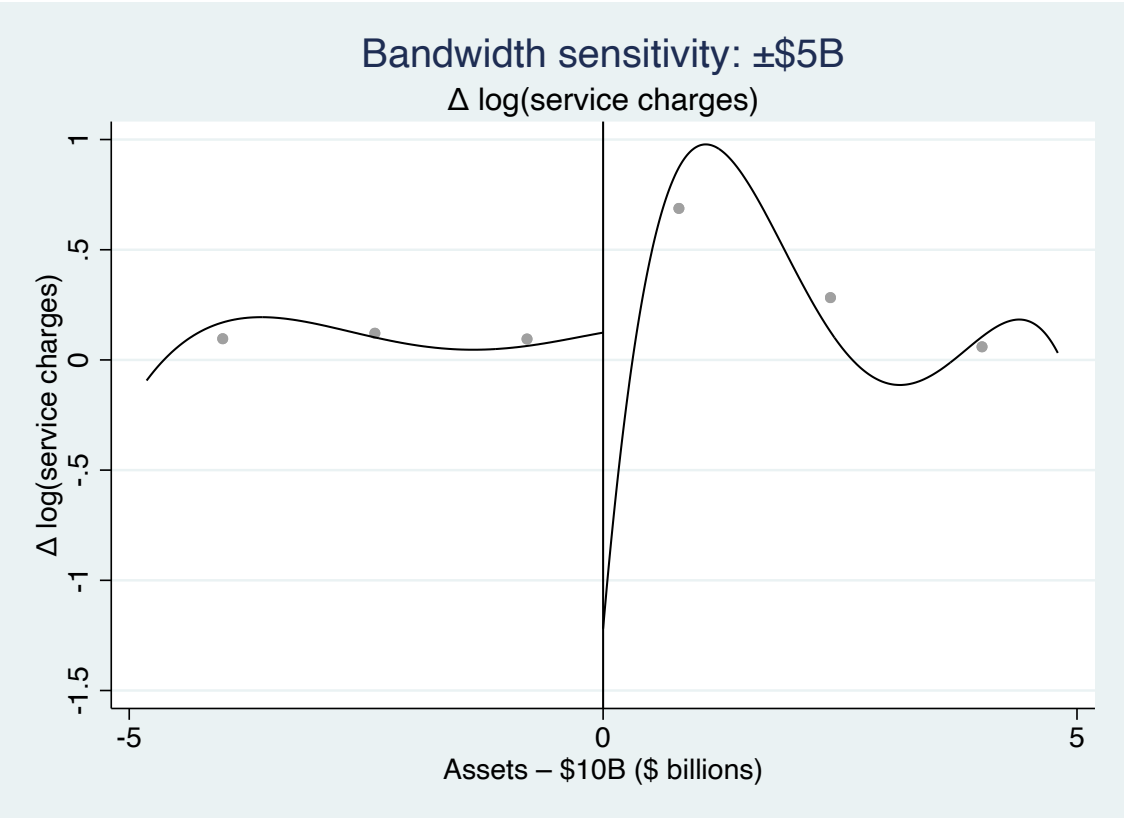

Figure 4: Bandwidth sensitivity of the RDD estimates for deposit fee changes: window of ±$5 billion around the cutoff.

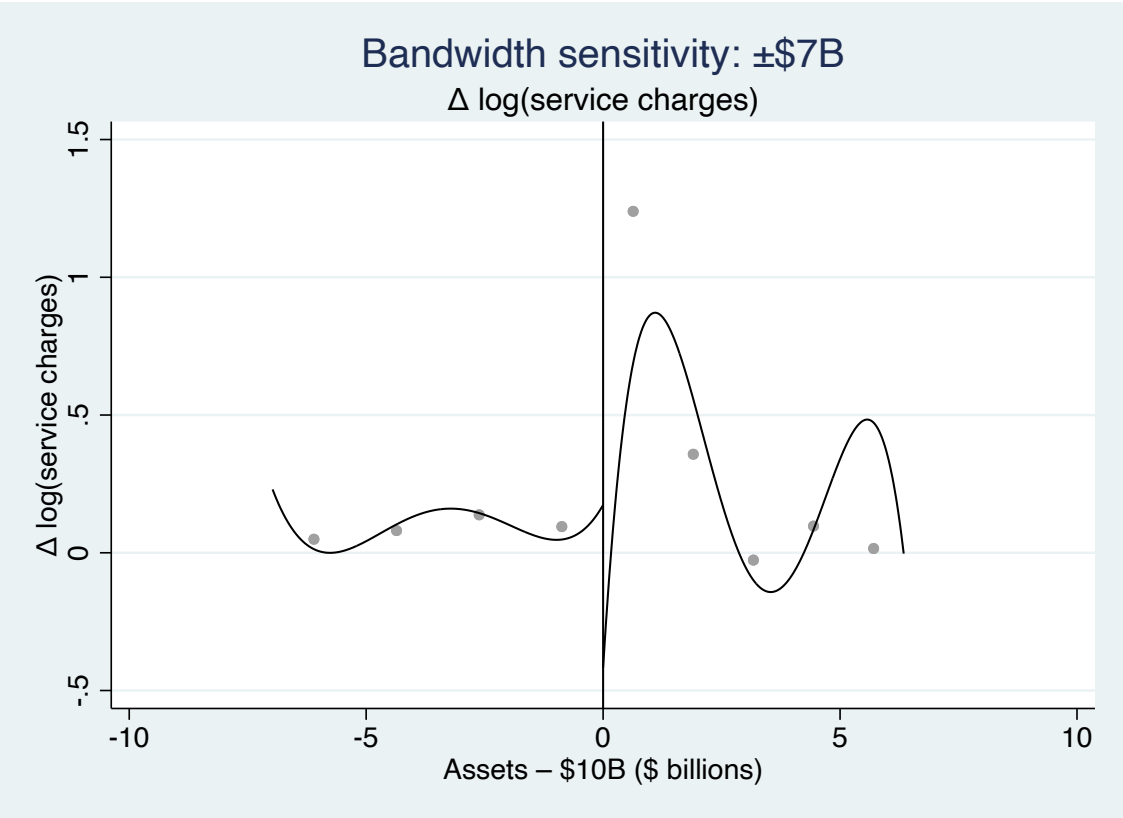

Figure 5: Bandwidth sensitivity of the RDD estimates for deposit fee changes: window of ±$7 billion around the cutoff.

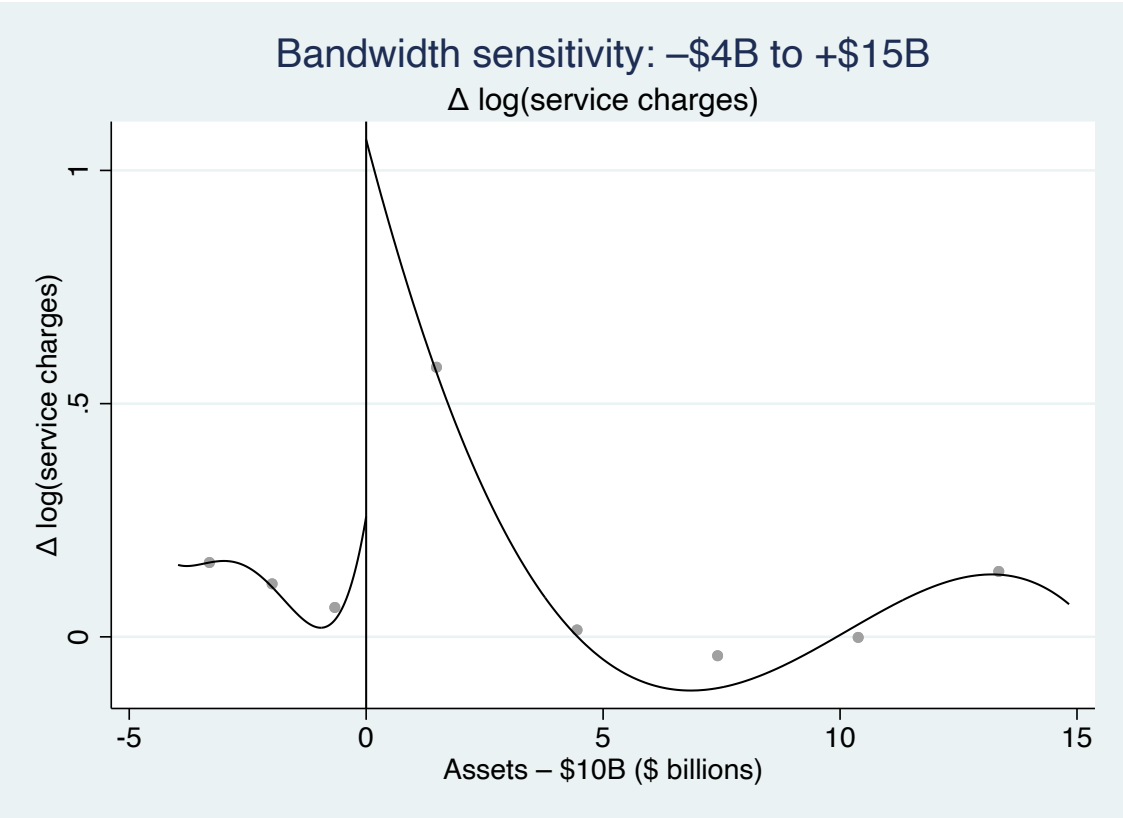

Figure 6: Bandwidth sensitivity of the RDD estimates for deposit fee changes: asymmetric window from −$4 billion to +$15 billion.

billion window the interval from -1.274 to 2.961 log points includes both large negative and very large positive effects.

The bottom rows of Table 3 and the sensitivity plots in Figures 4–6 show why the bandwidth results are so unstable. Even with the widest of these local windows, the effective number of banks above the cutoff ranges only from 12 to 22. In the strictest ±$5 billion window, there are just 12 treated banks and 29 control banks. With so few observations on the right side of the cutoff, small changes in which banks are included can cause large swings in the estimated jump, and the standard errors remain large even when the bandwidth is widened.

In short, the bandwidth sensitivity checks confirm that the RDD design is severely underpowered in this setting. The positive point estimates are consistent with substantial Durbin-induced fee increases, but the data cannot distinguish these from zero.

III.4 Covariate Balance and Density Diagnostics

Table 4: Covariate Balance at the $10 Billion Threshold (Pre-Policy Loans-to-Assets)

	Loans-to-assets ratio in 2011:Q2
RD estimate at $10B cutoff	-0.113
Std. error (approx.)	(0.054)
p-value	0.035
Approx. 95% confidence interval	$[-0.218,\ -0.008]$
Bandwidth (in $ billions)	50
Observations (banks)	3,851
Effective obs. below cutoff	3,837
Effective obs. above cutoff	13

Notes: Outcome is the pre-policy loans-to-assets ratio measured in 2011:Q2. The specification matches the baseline RDD setup but replaces the outcome with this pre-treatment covariate, providing a test of covariate balance at the $10 billion cutoff. Estimates are from `rdrobust` with a local linear specification, triangular kernel, and HC1 robust variance. The standard error and confidence interval are rounded, based on the reported estimate and z-statistic.

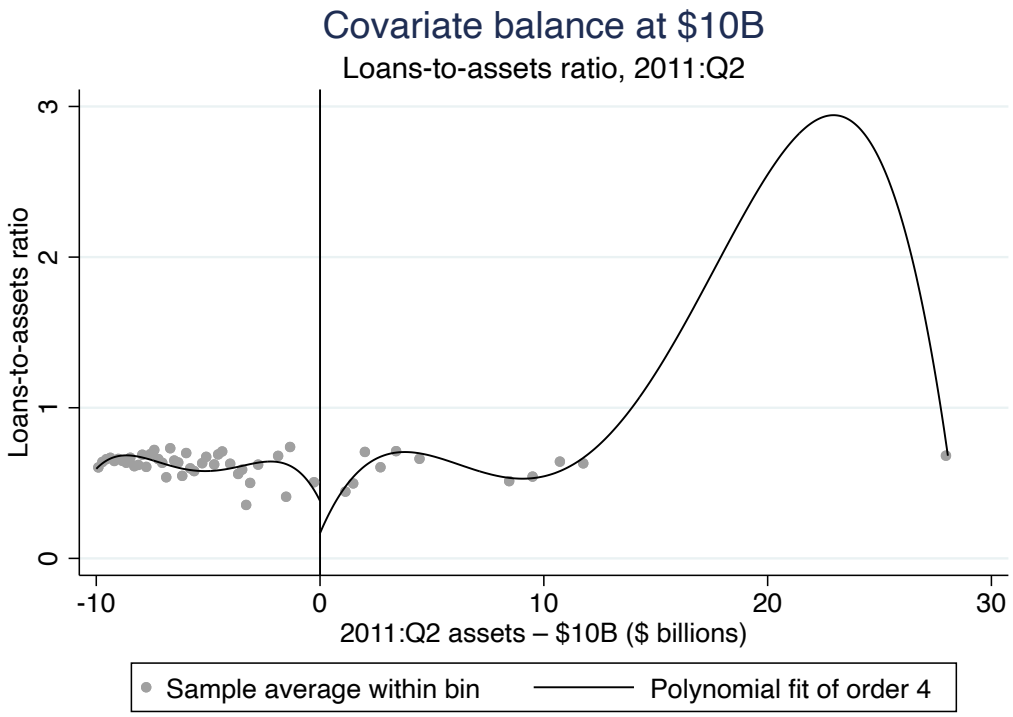

Figure 7: Covariate balance in pre-policy loans-to-assets ratios at the $10 billion cutoff. Binned scatter of 2011:Q2 loans-to-assets versus 2011:Q2 assets relative to $10 billion, with separate local polynomial fits.

Table 4 summarizes the covariate-balance test at the $10 billion threshold, using the loans-to-assets ratio in 2011:Q2 as the outcome and keeping the same RDD specification as in the baseline model. The estimated jump in this pre-policy covariate at the cutoff is -0.113, with an approximate standard error of 0.054 and a p-value of 0.035. This means that banks just above $10 billion have loans-to-assets ratios that are about 11 percentage points lower, on average, than banks just below the threshold. Figure 7 shows a downward step at the cutoff that matches this estimate.

This result is not ideal for identification. In a perfectly balanced RDD, pre-policy covariates would evolve smoothly through the threshold. Here, the significant intercept shift suggests that banks above and below $10 billion differ in their pre-Durbin business models in a way that is not fully captured by the running variable itself. Including loans-to-assets as a control partly addresses this concern by conditioning on the most visible imbalance, but it cannot guarantee balance on unobserved characteristics.

The density diagnostics based on Figure 1 tell a complementary story. The histogram shows a very steep cliff at $10 billion: there are thousands of small banks with assets far below the threshold but only a few dozen banks even moderately close to it. The density of banks is much lower above the cutoff than below. Importantly, there is no visual evidence of bunching right below the cutoff. The cliff is driven by the overall shape of the size distribution rather than by strategic manipulation.

This supports the no-manipulation assumption, but it also underscores the main weakness of the design: the effective treated group near $10 billion is tiny.

III.5 Interpretation and Comparison to Prior Work

Taken together, the results suggest that Durbin-covered banks likely raised deposit fees relative to uncovered banks, but the Call Report data do not allow a precise estimate of the size of this effect. The main RDD estimates are positive but statistically insignificant, and the confidence intervals remain wide even after adding a key pre-policy covariate. Bandwidth checks show that narrower windows around the cutoff generate very large point estimates, but these are driven by a handful of treated banks and are too imprecise for firm conclusions. Covariate and density diagnostics highlight the core limitation: there are very few mid-sized banks near the $10 billion threshold, and those that exist differ in observable characteristics such as loan intensity.

These findings are therefore best read as inconclusive but consistent with the large waterbed effects found in bank-holding-company data by Kay, Manuszak, and Vojtech (2018) and Mukharlyamov and Sarin (2025). The estimates neither provide strong evidence against substantial recoupment of interchange losses through deposit fees, nor pin down the magnitude of that recoupment. Instead, they show that aggregate service-charge income from commercial bank Call Reports, combined with the sparse distribution of banks near the $10 billion cutoff, leaves the sharp RDD underpowered as a tool for measuring Durbin's impact on deposit fee revenue.

Several features of the data and design limit what this study can say. First, the core issue is not classical sampling error but lack of support around the threshold (Lee and Lemieux, 2010). The Federal Reserve's issuer lists identify 9,291 banks and thrifts as debit-card issuers. After merging those lists to the public Call Reports and FDIC data, the analysis recovers usable data for about 3,970 banks, of which 3,905 enter the final RDD sample. Many institutions cannot be matched reliably or are missing key variables, such as 2011:Q2 assets or service charge income, in the public supervisory files, so they drop out before the RDD is constructed. This mechanical data loss is a major reason why only 42 banks lie above $10 billion and why very few control banks sit close to the cutoff.

Second, measurement matters. Call Report item RIAD4080 aggregates a wide variety of service charges, including overdraft fees, ATM fees, account maintenance fees, and other charges on deposit accounts. It does not isolate the specific fees most directly tied to debit card usage or to Durbin-motivated account restructuring. Banks could respond to Durbin by shifting revenue into other line items, altering interest margins, or adjusting non-price dimensions such as branch networks and service quality. If much of the waterbed effect operates through these other channels, then even a large underlying change in the total cost of banking services to consumers could appear as a small or noisy shift in this one aggregate fee category.

Third, this bank-level sample is not the same population as the entities analyzed in prior work. Studies such as Kay, Manuszak, and Vojtech (2018), Manuszak and Wozniak (2017), and Mukharlyamov

and Sarin (2025) use bank-holding-company data or proprietary account-level fee schedules that capture consolidated operations of large national banking groups. By contrast, the sample here includes only insured commercial banks that can be matched to the Federal Reserve's issuer lists and have non-missing assets in 2011:Q2. This design over-represents very small community banks and under-represents mid-sized and large consolidated organizations.

Fourth, there are plausible economic reasons why a true Durbin effect on deposit fees might be small or hard to detect at the cutoff, even aside from power. Retail banking is highly concentrated, and large banks often act as price leaders in local markets. If covered banks raised their fees in response to Durbin, smaller exempt banks may have partially matched those increases to maintain margins or avoid signaling lower quality. Such spillovers would compress fee differences across the threshold and attenuate any discontinuity at $10 billion, even though the policy still shifted costs onto depositors in absolute terms.

Set against the existing literature, these findings are best interpreted as a cautionary note rather than a contradiction. Account-level and holding-company studies typically find sizable pass-through of Durbin's interchange cap into higher consumer fees, often on the order of 70 to 90 percent of lost interchange revenue. The point estimates here are smaller and imprecise, and the confidence intervals are wide enough that those magnitudes cannot be statistically rejected. The difference in conclusions reflects differences in data, units of analysis, and the density of observations near the threshold, rather than clear evidence that Durbin had no impact on deposit fees.

IV. Conclusion

This paper asks whether the Durbin Amendment's cap on debit interchange fees led covered banks to raise deposit service charges more than similar uncovered banks. Using FFIEC Call Report data from 2009–2013 and the Federal Reserve's issuer lists, it estimates a sharp RDD centered at the policy's $10 billion asset threshold. The main RDD estimates, using a wide manual bandwidth of ±$50 billion in 2011:Q2 assets, show a positive but statistically insignificant jump in fee growth at the threshold, about 0.21 log points without covariates and 0.42 with the pre-policy loans-to-assets ratio. None of the narrower symmetric or asymmetric bandwidths delivers a precise or robust effect. Taken together, the evidence does not reveal a clear discontinuity in deposit fee changes at $10 billion.

What follows from this pattern? First, the local treatment effect at the cutoff, if it exists, is estimated with very low precision. The confidence intervals around the RDD estimates are wide enough to include both economically meaningful waterbed effects and true zero effects. Second, adding a pre-policy loans-to-assets ratio improves precision slightly and confirms that business-model characteristics predict fee changes, but it does not turn the Durbin effect into a statistically significant result. Third, the event-study specification using the full panel shows significant pre-trends for large banks in the two quarters immediately before Durbin, which undermines a standard difference-in-differences design and further justifies focusing on the cutoff-based RDD.

Several features of the data and design limit the scope of these findings. The main constraint is the sparse distribution of banks near the $10 billion threshold, which leaves the RDD severely underpowered. A second limitation is the coarse outcome measure: a single aggregated line item for service charges on deposit accounts that may miss important margins of adjustment in fees and non-price terms. A third limitation is that the bank-level sample does not align perfectly with the population of economic decision-makers analyzed in prior work, which often focuses on holding companies or specific account types.

In the broader context, this study underscores how difficult it is to learn about the incidence of financial regulation from public supervisory aggregates alone. For the motivating question, whether banks offset Durbin's interchange cap by raising deposit fees for ordinary consumers, the answer from these data must remain modest. The results do not provide sharp evidence of a discrete jump in deposit service charges at the $10 billion threshold, but they also do not rule out sizable waterbed effects. Instead, they highlight two broader lessons. First, quasi-experimental identification in banking often hinges on having enough institutions at the relevant regulatory margin. Second, banks' responses to regulation may be spread across multiple products, customer segments, and organizational levels in ways that a single Call Report line item cannot capture.

Future work that combines the RDD logic with richer micro-data, such as account-level fees, local market structure, and holding-company balance sheets, would be better positioned to quantify how and where consumers ultimately pay for debit interchange regulation.

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